

Advanced Natural Language Processing

Discussion 08: XAI for NLP

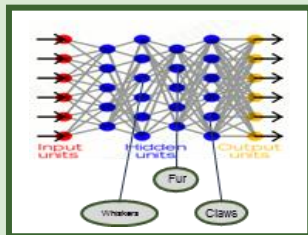
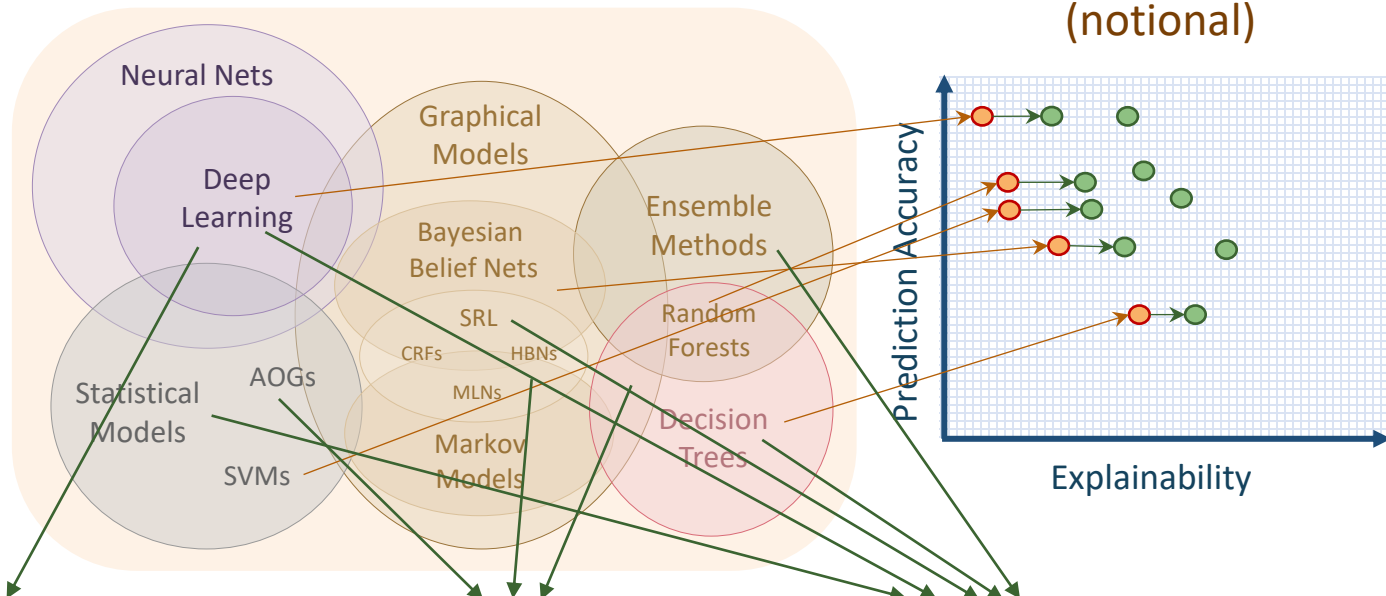
Department of Advanced Computer Sciences (DACS)
Master of AI / DSDM
Prof dr ir Jan C. Scholtes – Course year 2026-2027



Maastricht University

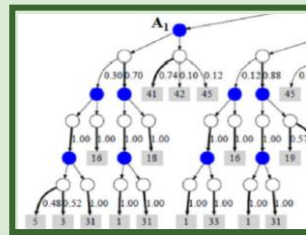
Learning Techniques (today)

Explainability (notional)



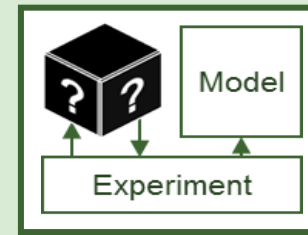
Deep Explanation

Modified deep learning techniques to learn explainable features



Interpretable Models

Techniques to learn more structured, interpretable, causal models



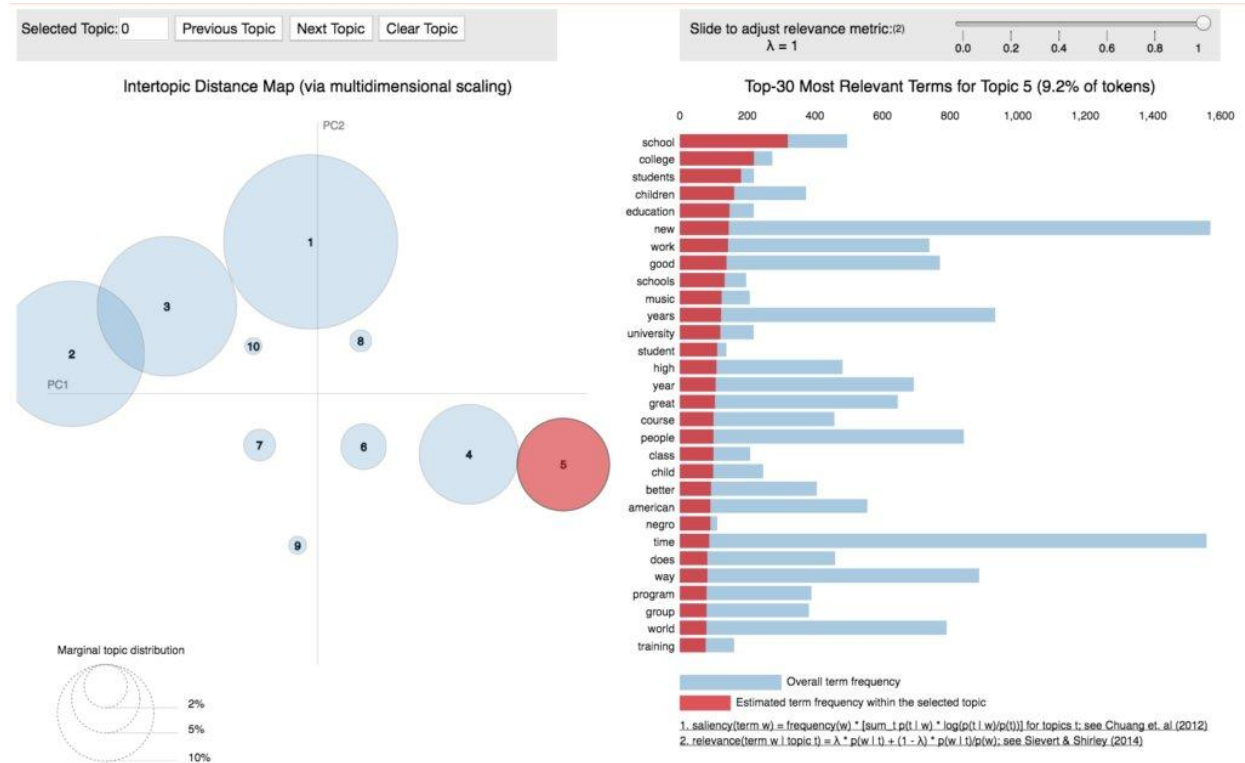
Model Induction

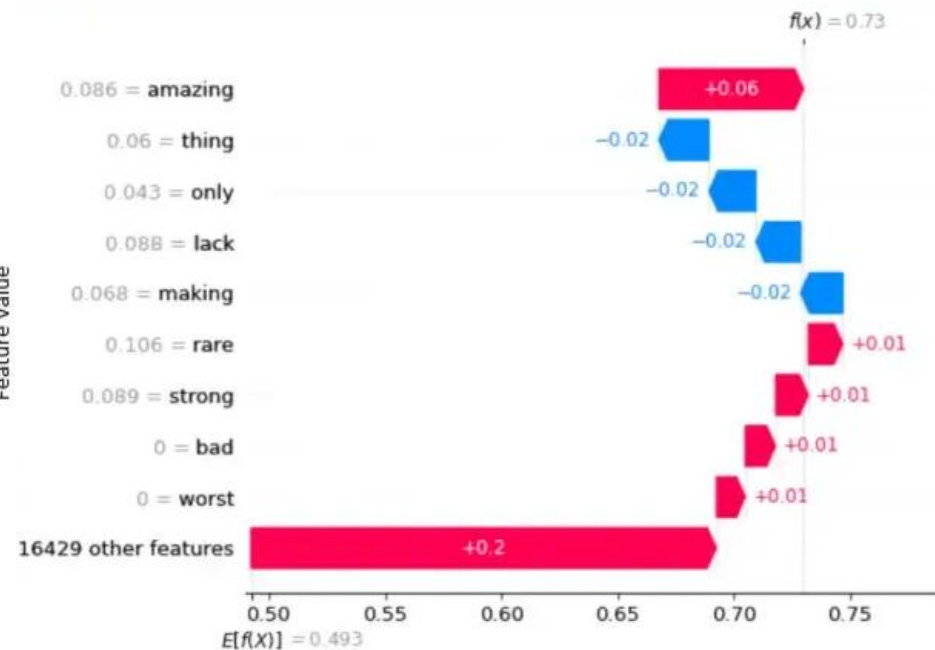
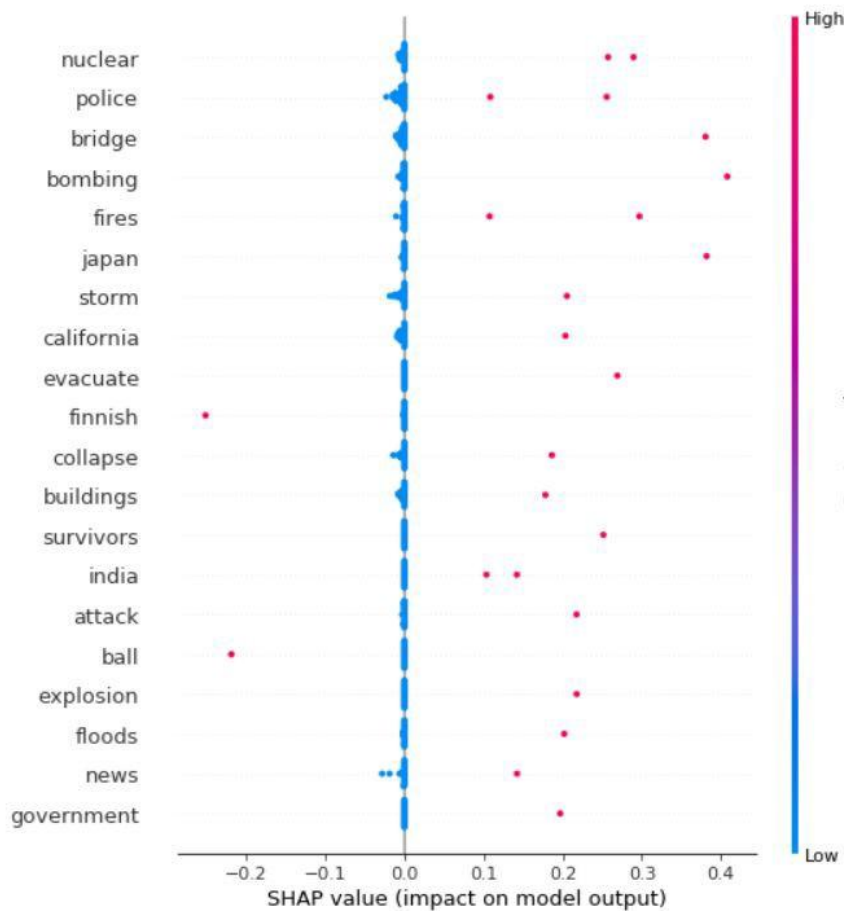
Techniques to infer an explainable model from any model as a black box

Source: Darpa

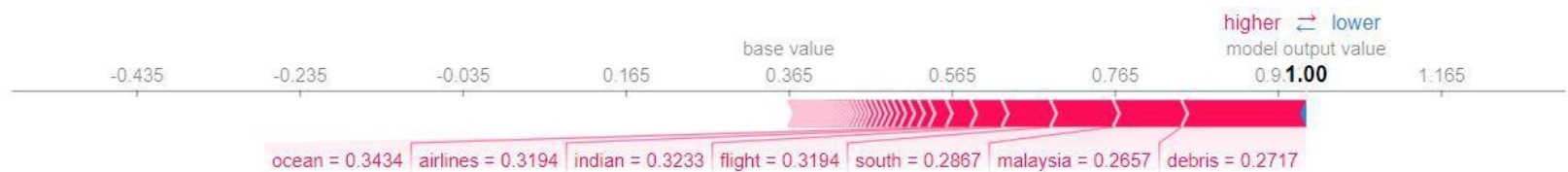
LDAViz for LDA and NMF

- Python library for interactive topic model visualization.
- Also has implementations for NMF.
- Shows topics and dominant keywords per topic.
- Standard Python implementations.



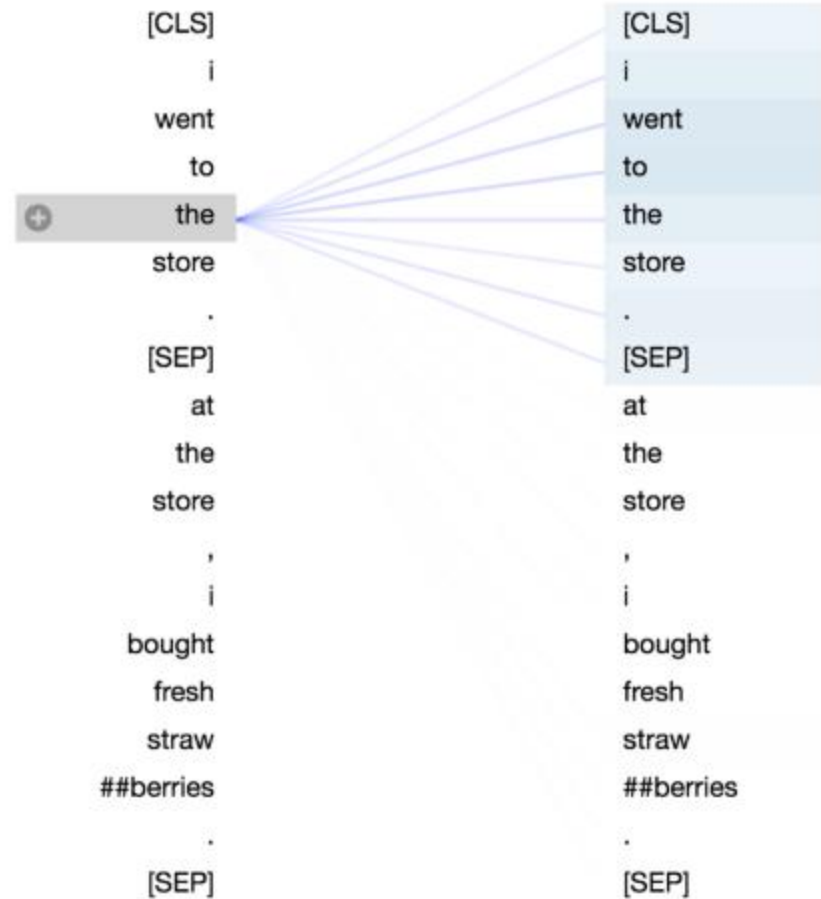


Source: <https://medium.com/nlplanet/two-minutes-nlp-explain-predictions-with-shap-values-2a0e34219177>

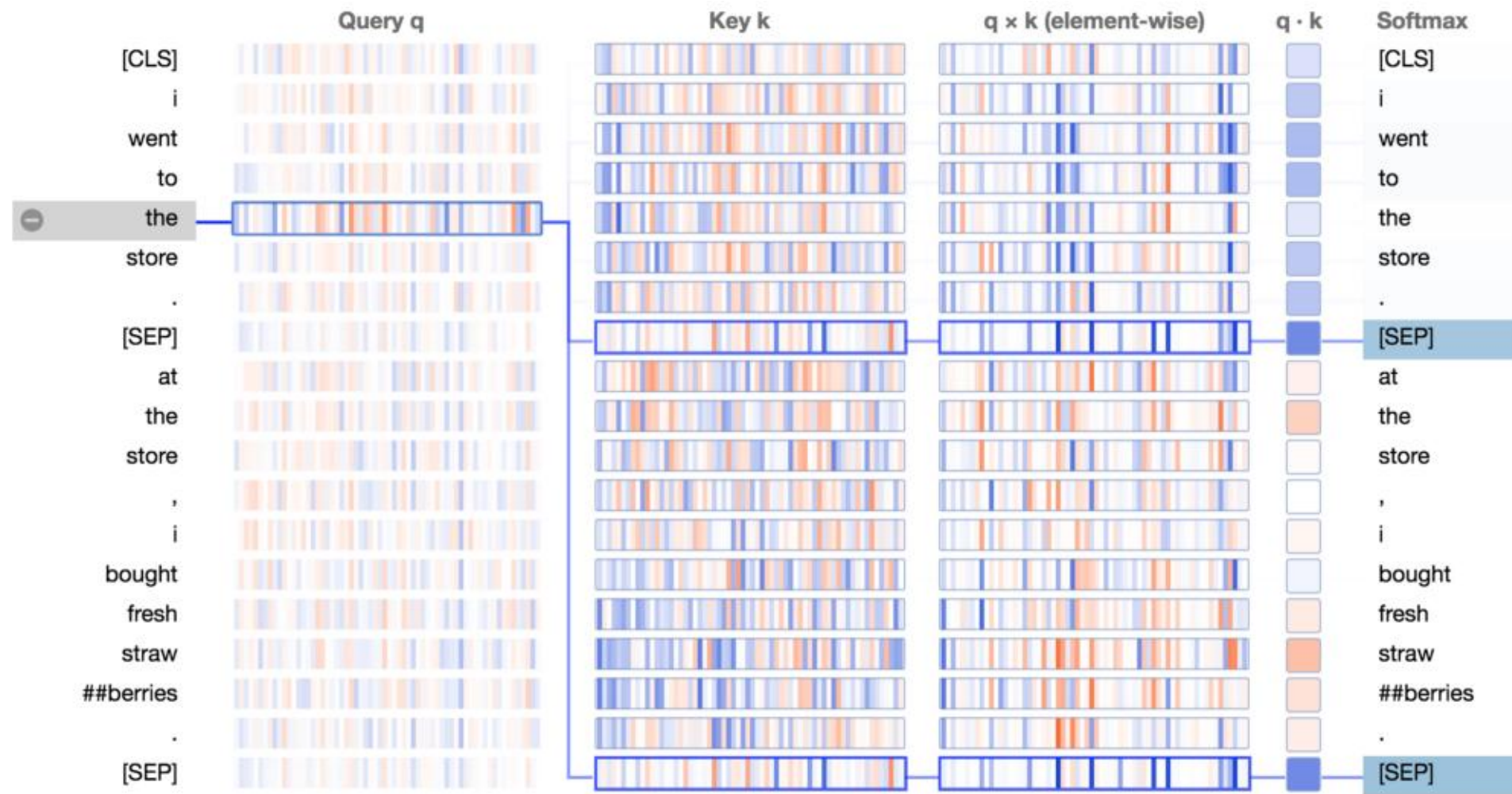


See also: Lundberg, S. (2018). SHAP. MIT. [Python package]. Retrieved from <https://github.com/slundberg/shap>

BERTViz: Head View

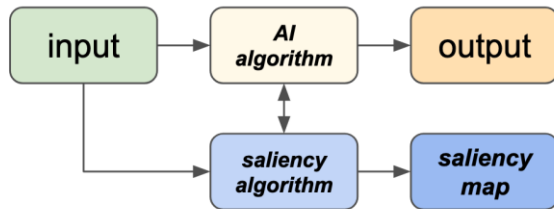


BERTViz: Neuron View

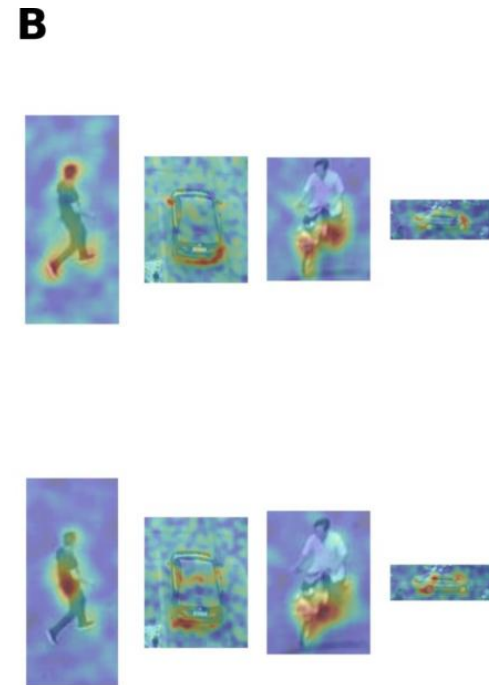
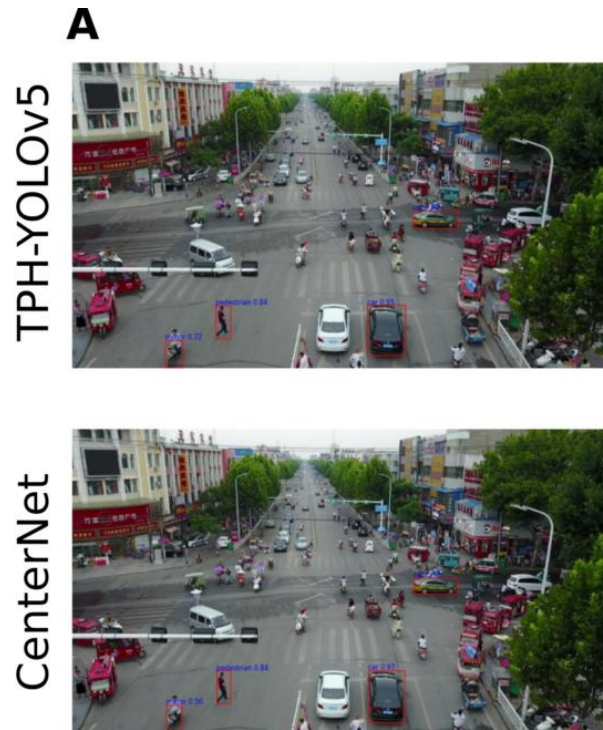


Saliency

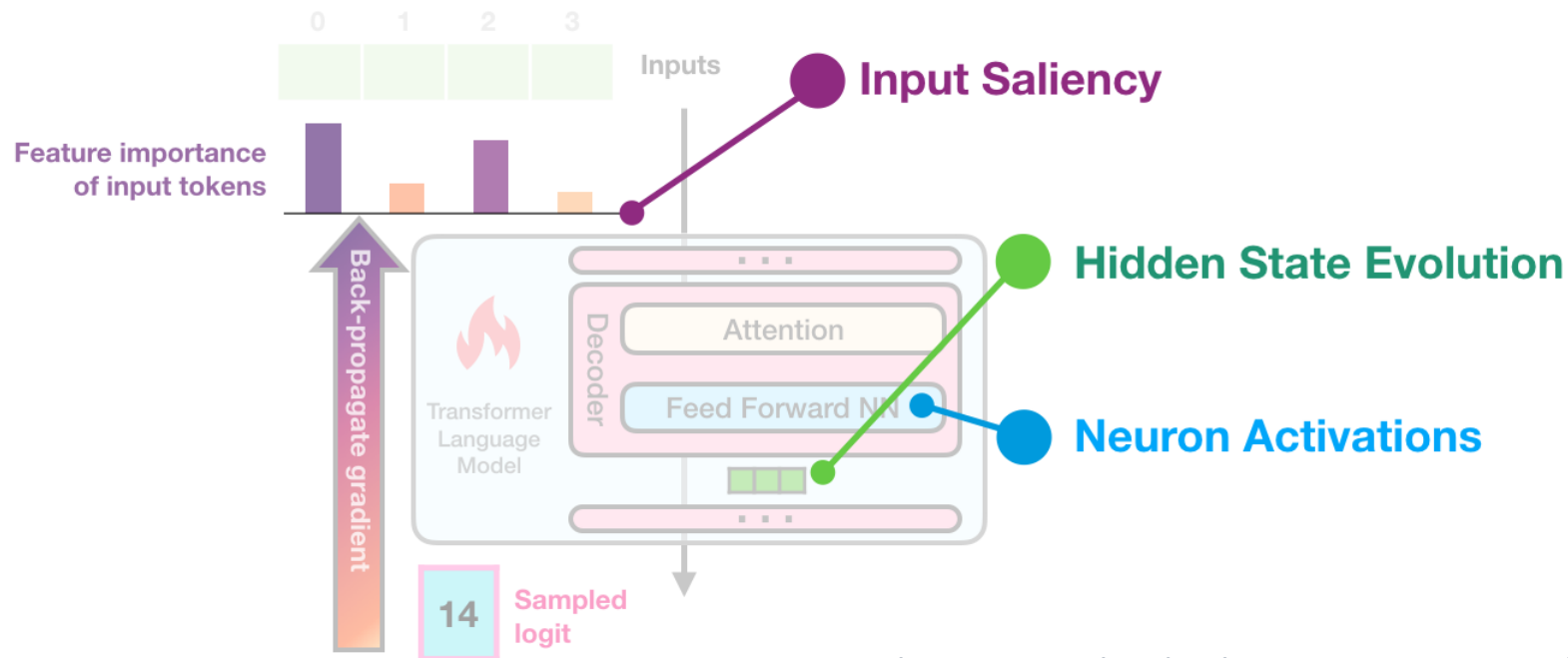
- Saliency maps are heat maps that highlight features in the input to an AI model that were significant in the AI model's output predictions.
- Originates from computer vision with deep learning.



(A) Both models produce similar high-confidence detections (shown with red bounding boxes). **(B)** Despite having similar detections, the saliency maps corresponding to these detections reveal subtle differences in the input features used by the two models, e.g. TPH-YOLOv5 (top) focuses on the head and feet of pedestrians, while CenterNet (bottom) focuses on the torso.



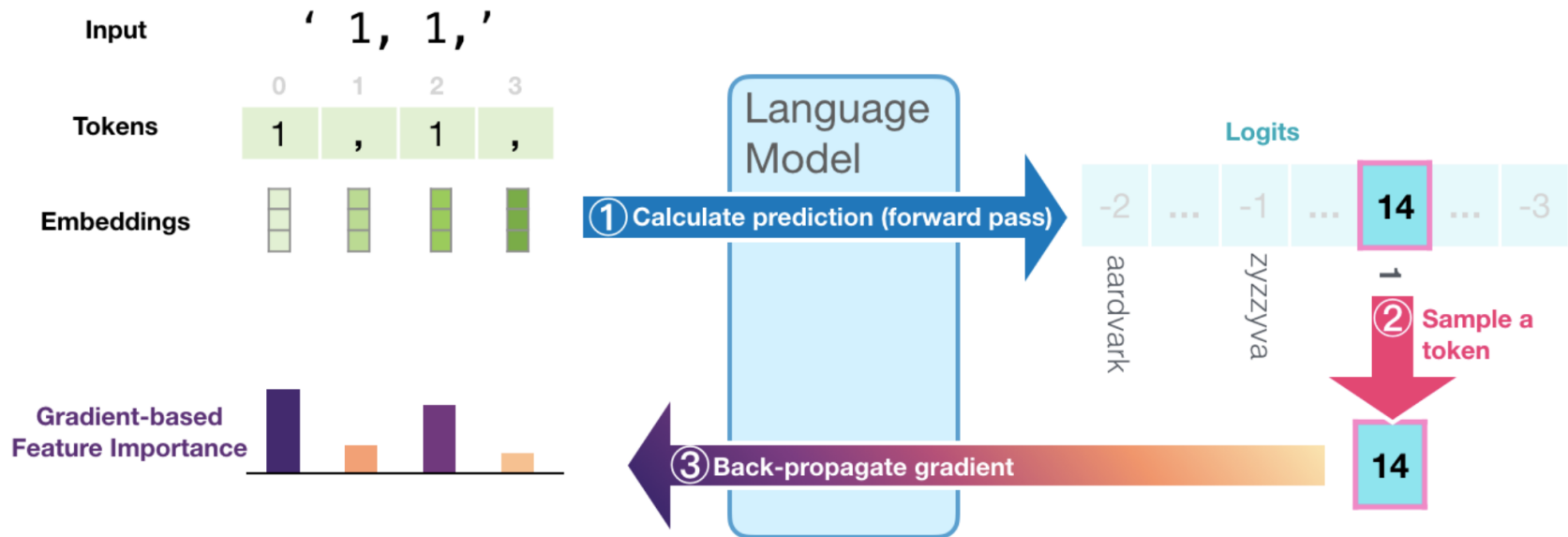
Saliency Methods for Neural Language Models



Source: <https://jalammar.github.io/explaining-transformers/>

- Input Saliency methods that score input tokens importance to the output. Can be used for classification as well as for generation.
- Neuron Activations and how individual and groups of model neurons spike in response to inputs and to produce outputs.
- Hidden State Evolution across the layers of the model and what it may tell us about each layer's role.

Gradient-based Saliency



1. The language model's final hidden state is projected into the model's vocabulary resulting in a numeric score for each token in the model's vocabulary. Using a softmax operation results in a probability score for each token.
2. We proceed to select a token (e.g. select the highest-probability scoring token or another strategy as we discussed).
3. By calculating the gradient of the selected logit (before the softmax) with respect to the inputs by back-propagating it all the way back to the input tokens, we get a signal of how important each token was in the calculation resulting in this generated token. That assumption is based on the idea that the smallest change in the input token with the highest feature-importance value makes a large change in what the resulting output of the model would be.

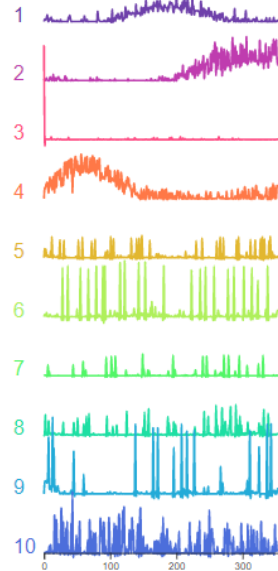
Non-Negative Matrix Factorization (NMF)

- Non-Negative Matrix Factorization (NMF) is a math technique that splits a large grid of positive numbers into two smaller positive grids that, when multiplied together, recreate the original data.
- Think of it like a recipe: it takes a finished dish (your main data) and breaks it back down into the individual ingredients (the building blocks) and the amounts used (the weights).
- We discuss this in detail in the course Agentic Conversational Search.

$$\begin{bmatrix} & \\ & \\ & \\ & \end{bmatrix}^W \times \begin{bmatrix} & & & & & \\ & & & & & \end{bmatrix}^H \approx \begin{bmatrix} & & & & & \\ & & & & & \\ & & & & & \\ & & & & & \end{bmatrix}^V$$

Neuron Activations

- Feed Forward Neural Network (FFNN) is one of the main components of a transformer on top of the self-attention.
- It contains 66% of all parameters, so it holds a significant part of the model's capacity.
- What happens where?
- Using Non-negative Matrix Factorization (NMF) for reducing the dimensionality into groups that are potentially individually more interpretable.



Now I ask you : what can be expected of man since he is a being endowed with strange qualities ? Shower upon him every earthly blessing , drown him in a sea of happiness , so that nothing but bubbles of bliss can be seen on the surface ; give him economic prosperity , such that he should have nothing else to do but sleep , eat cakes and busy himself with the continuation of his species , and even then out of sheer ingratitude , sheer spite , man would play you some nasty trick . He would even risk his cakes and would deliberately desire the most fatal rubbish , the most uneconomical absurdity , simply to introduce into all this positive good sense his fatal fantastic element . It is just his fantastic dreams , his vulgar folly that he will desire to retain , simply in order to prove to himself -- as though that were so necessary -- that men still are men and not the keys of a piano , which the laws of nature threaten to control so completely that soon one will be able to desire nothing but by the calendar . And that is not all : even if man really were nothing but a piano - key , even if this were proved to him by natural science and mathematics , even then he would not become reasonable , but would purposely do something perverse out of simple ingratitude , simply to gain his point . And if he does not find means he will contrive destruction and chaos , will contrive sufferings of all sorts , only to gain his point ! He will launch a curse upon the world , and as only man can curse (it is his privilege , the primary distinction between him and other animals) , may be by his curse alone he will attain his object -- that is , convince himself that he is a man and not a piano - key ! \n

>> Well

Explorable: Ten Factors From The Underground

Tap or hover over the sparklines on the left to isolate a certain factor

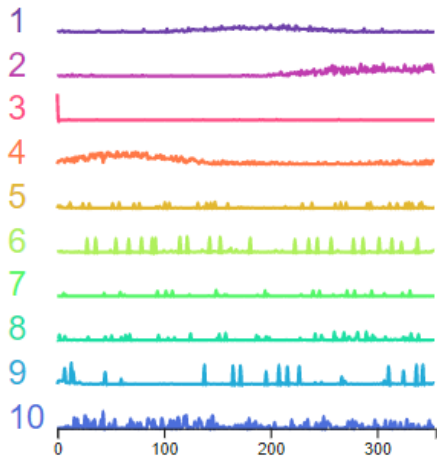
Ten Factors from the activations the neurons in Layer 0 in response to a passage from Notes from Underground by Dostoevsky.

- Factors that focus on specific portions of the text (beginning, middle, and end). This is interesting as the only signal the model gets about how these tokens are connected are the positional encodings. This indicates the neurons that are keeping track of the order of words. Further examination is required to assess whether these FFNN neurons directly respond to the time signal, or if they are responding to a specific trigger in the self-attention layer's transformation of the hidden state.
- Factors corresponding to linguistic features, for example, we can see factors for pronouns (he, him), auxiliary verbs (would, will), other verbs (introduce, prove -- notice the understandable misfiring at the 'suffer' token which is a partial token of 'sufferings', a noun), linking verbs (is, are, were), and a factor that favors nouns and their adjectives ("fatal rubbish", "vulgar folly").
- A factor corresponding to commas, a syntactic feature.
- The first-token factor

Go here: for a live view of what happens in the different layers: <https://jalammar.github.io/explaining-transformers/>

Try this all yourself with ECCO

```
[ ] nmf_2.explore()
```



Now I ask you : what can be expected of man since he is a being endowed with strange qualities ? Shower upon him every earthly blessing , drown him in a sea of happiness , so that nothing but bubbles of bliss can be seen on the surface ; give him economic prosperity , such that he should have nothing else to do but sleep , eat cakes and busy himself with the continuation of his species , and even then out of sheer ingratitude , sheer spite , man would play you some nasty trick . He would even risk his cakes and would deliberately desire the most fatal rubbish , the most uneconomical absurdity , simply to introduce into all this positive good sense his fatal fantastic element . It is just his fantastic dreams , his vulgar folly that he will desire to retain , simply in order to prove to himself -- as though that were so necessary -- that men still are men and not the keys of a piano , which the laws of nature threaten to control so completely that soon one will be able to desire nothing but by the calendar . And that is not all : even if man really were nothing but a piano - key , even if this were proved to him by natural science and mathematics , even then he would not become reasonable , but would purposely do something perverse out of simple ingratitude , simply to gain his point . And if he does not find means he will contrive destruction and chaos , will contrive sufferings of all sorts , only to gain his point ! He will launch a curse upon the world , and as only man can curse (it is his privilege , the primary distinction between him and other animals) , may be by his curse alone he will attain his object -- that is , convince himself that he is a man and not a piano - key ! \n

>> This

- https://colab.research.google.com/github/jalammar/ecco/blob/main/notebooks/Ecco_Neuron_Factors.ipynb#scrollTo=CLDsArU2N0lr



Jacobian Matrix & J-Space

- **Jacobian matrix:** In calculus, a Jacobian matrix is the matrix of all first-order partial derivatives of a vector-valued function. It describes how small changes in a system's inputs translate into changes in its outputs.
- **Application to Claude:** The J-Lens asks, for any internal activation inside the model, “if this activation were nudged slightly, how would that change the probability that the model outputs each possible word later on?” This sensitivity relationship is mathematically a Jacobian — hence the name “Jacobian Lens.”
- **How it is computed:** For every layer ℓ of Claude's residual stream, the J-Lens computes the averaged linearized effect of an activation on the model's final output logits: formally, $J_\ell = E[\partial h_{\text{final}} / \partial h_\ell]$, averaged over roughly 1,000 diverse prompts and token positions. Applying this averaged Jacobian to any activation at any layer produces a ranked list of vocabulary words — the words the model is internally “oriented toward” at that moment, whether or not it ever says them. The paper reports that a typical J-Space holds only 10–25 active J-Lens vectors at once, together accounting for less than 10% of a given representation's activation variance.

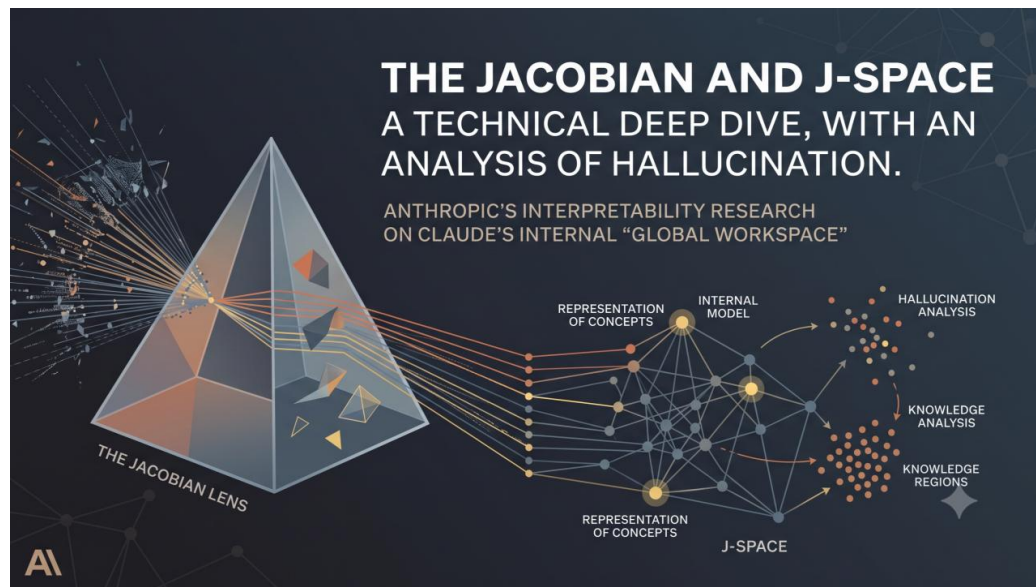
Example: When Claude reads a snippet of code containing a bug nobody has flagged, applying the J-Lens to its internal state reveals the token “ERROR” rising to the top of the list — even though Claude's visible output says nothing about a bug yet.

Recent Development: The Jacobian Lens and “J-Space” (2026)

- The problem: the Logit Lens (applying the final unembedding matrix to a middle layer) ignores the non-linear transformations still to come, giving an inaccurate read of what a middle layer “means”.
- The fix: Anthropic’s Jacobian lens (J-lens) computes the Jacobian of the remaining layers with respect to a middle layer — a calculus-based “fast-forward” that projects hidden activations into vocabulary space far more accurately than the Logit Lens.
- J-space: applying the J-lens to Claude Opus 4.6 revealed a privileged internal workspace holding concepts, multi-step reasoning, and even hidden intentions (e.g. recognizing it is being tested) — running in parallel to, and structurally separate from, the track that generates the visible output.

Why it belongs here: a direct, gradient-based successor to today’s saliency and hidden-state methods — concrete evidence that attention weights alone cannot explain what a model is “thinking”.

Gurnee, Lindsey et al., “Verbalizable Representations Form a Global Workspace in Language Models”, transformer-circuits.pub, July 2026. See also: [MIT Technology Review coverage](#)



Any Questions on the Lecture

Now let's form groups of 4



Discuss Exercises Lecture

Exercise 1: Model-Agnostic Post-Hoc Interpretability (LIME vs. SHAP)

The slides examine post-hoc local explanation techniques that treat the NLP model as a black box. Consider a sentiment classification model predicting whether online product reviews are negative or non-negative.

- Explain how **LIME** (Local Interpretable Model-agnostic Explanations) creates local explanations for a single review. How does it perturb input text, obtain predictions from the black-box model, and fit an interpretable surrogate model weighted by proximity?

In the lecture, **Shapley values** from cooperative game theory:

- $\phi_i(v) = \frac{1}{|N|!} \sum_R [v(P_i^R \cup \{i\}) - v(P_i^R)]$
- Using the 3-chef analogy from slides 12 and 13, explain what $v(P_i^R \cup \{i\}) - v(P_i^R)$ represents for a specific feature (or token) i . Why does calculating exact Shapley values require computing all $|N|!$ permutations of features, and why does this become computationally prohibitive for long textual sequences?
- The example sentence *"about to go shopping again tomorrow bc the dress i got for jason aldean is not cute"* is explained for the negative class by the weights on *"not"* and *"cute"*. What is the primary limitation of token-attribution methods when explaining non-linear semantic interactions like negation or sarcasm?

Exercise 2: White-Box Interpretability (Attention Visualizations vs. Gradient Saliency)

The slides contrast visual inspection of internal attention weights (BERTViz) with gradient-based saliency methods.

- **BERTViz** distinguish between the **Head View** and the **Neuron View**. What specific information does the neuron view provide regarding the element-wise interaction ($q \times k$) and the subsequent softmax attention weight before multiplying by value vectors?
- We also illustrate **Gradient-based Saliency**. Explain how backpropagating the gradient of the chosen unnormalized logit score (before softmax) back to the input token embeddings provides a feature importance score. What core mathematical assumption connects the magnitude of $\frac{\partial \text{logit}}{\partial x_i}$ to the impact of that token on the final output?
- Based on the debate cited in earlier lectures ("*Attention is not Explanation*" vs. "*Attention is not not Explanation*"), explain why inspecting raw attention matrices can produce misleading explanations of model decisions, whereas gradient-based saliency directly measures sensitivity to changes in the input representation.

Exercise 3: Internal Representations (Hidden State Evolution and Neuron Factorization)

The lecture slides outline techniques for peering inside Transformer layers beyond simple input-attribution maps, focusing on Feed-Forward Network (FFN) activations and hidden states.

- We show that the position-wise Feed-Forward Networks (FFNs) contain approximately 66% of all parameters in a standard Transformer block. Why does analyzing self-attention alone fail to capture the majority of the model's stored capacity and non-linear transformations?
- How does the **ECCO** framework utilize **Non-negative Matrix Factorization (NMF)** on neuron activations across hidden layers to extract interpretable latent factors? What do distinct activation factors reveal about how different layers process low-level syntax versus abstract semantic properties?
- Explain the concept of **Hidden State Evolution** across the depth of a decoder or encoder stack. If an output token's top logit changes dynamically from layer 1 to layer 12, what does this tell an engineer debugging a model failure (such as hallucination or shortcut learning)?

Discussion



On Faithfulness and Plausibility: Can an Explanation Be Too Convincing?

A good explanation, for a human, is one that is sound, convincing, selective and builds trust (slide 7). A faithful explanation is one that reflects what the model actually computed (slide 6). LIME is built to satisfy the first: a simple linear story around one data point.

If an explanation is engineered to be believable, what is it evidence of: the model's reasoning, or our willingness to accept a story? Can an explanation be too good?

When a tool optimises for human acceptance, at what point does interpretability turn into self-deception, and who is responsible when a convincing explanation of a wrong decision makes everyone stop looking?

On Explaining Ourselves: Do Humans Meet the Standard We Set for Models?

Psychology has known for fifty years that humans do not have faithful access to the causes of their own decisions. Asked why, we produce a plausible story after the fact, and we believe it (Nisbett and Wilson, 1977). Every human explanation is, in the lecture's terms, plausible and of unknown faithfulness.

We accept these explanations from doctors, judges and each other every day. Why, then, do we demand faithfulness from a model? Is the demand unfair, or is a model the first decision-maker whose explanations could actually be faithful, so that for once we can insist?

If a model's explanation were perfectly faithful and nobody could understand it, would it be an explanation at all? Explanation is something done for someone: what does that make faithfulness worth?

On the Streetlight: Is There a Level at Which a Model Can Be Understood?

Explainability tools look at attention because attention can be drawn as lines between words. Two thirds of the model, the feed-forward layers, cannot be drawn and are mostly not looked at. The drunk searches for his keys under the streetlight because that is where the light is.

Suppose we built the light: factorised every neuron, traced every circuit, projected every hidden state (slides 20 to 24). Would a complete account of a trillion parameters be understanding, or a second object as opaque as the first?

Is there a level of description at which a large language model is understandable at all, and if that level is itself a simplified model of the model, what have we gained over the plausible story we started with?

Questions?



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Break

Let's work on the tutorials



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